- **Physical activity vs Moca correlation test**

Metric	Spearman Rho	P_value (FDR corrected)
Total_movement	0.075	0.0032361
Day_movement	0.068	0.005403
Night_movement	0.064	0.006539
Physical_activity_num_days	0.060	0.008693
Average_daily_activity	0.041	0.068184
Average_active_calories_kcal	0.038	0.199898

  - **Age Stratification**
    - **40-54, 55-69, 70+**
    - **All statistical significant correlations are lost**
    - **Age could be a significant confounding variable to the physical activity and moca correlation suggesting the true correlation between moca and physical activity is less than the numbers**
  - **Diabetes status Stratification**
    - **0 statistically significant correlations**

- **Physical activity unimpaired vs impaired significant differences Mann Whittney U Test**
  - **Cutoff 26**

Metric	rank biserial (diff in prob)	P_value (FDR corrected)
Total_movement	-0.075	0.010
Physical_activity_num_days	-0.082	0.007
Active calories	-0.033	0.233
Daily steps	-0.021	0.412

  - **Cutoff 23**

Metric	rank biserial (diff in prob)	P_value (FDR corrected)
Total_movement	-0.051	0.337
Physical_activity_num_days	-0.092	0.044

Active calories	**-0.017**	**0.632**
Daily steps	**0.017**	**0.632**

- **Cutoff 24**

Metric	rank biserial (diff in prob)	P_value (FDR corrected)
Total_movement	**-0.065**	**0.087**
Physical_activity_num_days	**-0.081**	**0.045**
Active calories	**-0.013**	**0.807**
Daily steps	**0.0011**	**0.972**

- **Age Stratification**
  - **40-54, 55-69, 70+**
  - **Some statistical significant differences without age stratification are lost**
- **Diabetes status Stratification**
  - **Only 1 physical_activity_num_days statistically significant after FDR correction**
  - **Every group, lower moca score leads with less activity**
  - **For insulin dependent groups found opposite trend in rank biserial**
    - **Impaired participants at every cutoff show higher median and are significant before FDR**
    - **Cutoff 26 (impaired n≈135 vs unimpaired n≈103)**

metric	raw p	FDR p	rank-biserial
total_movement	0.2766	0.3688	+0.0824
physical_activity_num_days	0.4988	0.4988	−0.0507
average_active_calories_kcal	0.1518	0.3688	+0.1071
average_daily_activity	0.2357	0.3688	+0.0893

Cutoff 24 (impaired n≈84 vs unimpaired n≈154)

metric	raw p	FDR p	rank-biserial
total_movement	0.0610	0.1219	+0.1471
physical_activity_num_days	0.6857	0.6857	+0.0315
average_active_calories_kcal	0.1216	0.1622	+0.1196
average_daily_activity	**0.0230**	0.0919	**+0.1779**

Cutoff 23 (impaired n≈71 vs unimpaired n≈167)

metric	raw p	FDR p	rank-biserial
total_movement	0.0502	0.0669	+0.1606
physical_activity_num_days	0.4806	0.4806	+0.0574
average_active_calories_kcal	**0.0297**	0.0594	**+0.1757**
average_daily_activity	**0.0257**	0.0594	**+0.1824**

- **Age + Diabetes status Stratification**
  - **Statistically significant after FDR**

Age	Diabetes group	Cutoff	Metric	n (imp/unimp)	effect (rank-biserial)	raw p	FDR p
40–54	pre_diab	26	total_movement	57 / 128	**−0.2775**	0.002613	0.010450
40–54	pre_diab	26	average_active_calories_kcal	58 / 132	**−0.2136**	0.019201	0.038403
55–69	oral_med	26	physical_activity_num_days	134 / 170	**−0.2448**	0.000228	0.000914
55–69	oral_med	24	physical_activity_num_days	82 / 222	**−0.2587**	0.000500	0.002000
55–69	oral_med	23	physical_activity_num_days	56 / 248	**−0.2182**	0.010350	0.041400

  - **Statistically significant before FDR**

Part B — Suggestive (raw p<0.05, fails FDR)

The **insulin-dependent reversal shows up again** and is the standout suggestive pattern:

Cell	Cutoff	Metric	effect	raw p
55–69, insulin_dep	26	total_movement	**+0.245**	0.032
55–69, insulin_dep	26	daily_activity	+0.237	0.037
55–69, insulin_dep	24	total_movement	+0.245	0.034
55–69, insulin_dep	24	daily_activity	+0.283	0.014

- **2Sample 12 per strata of by wear_quality (sensor_sampling_duraiton_days)**

=== Glucose overall delta bins ===

by bin:

count pct

dup_0s	0	0.00
near_dup_0-10s	0	0.00
high_freq_10s-3min	0	0.00
near5_low_3-<5min	53	0.05
exact_5min	100261	99.85
near5_high_>5-7min	53	0.05
short_gap_7-30min	26	0.03
long_gap_>30min	15	0.01

=== Env overall delta bins ===

by bin:

	count	pct
dup_0s	0	0.0
near_dup_0-10s	6286205	100.0
high_freq_10s-3min	40	0.0
near5_low_3-<5min	1	0.0
exact_5min	0	0.0
near5_high_>5-7min	0	0.0
short_gap_7-30min	21	0.0
long_gap_>30min	13	0.0

=== Activity delta bins for point signals ===

signal	heart_rate	oxygen_saturation	physical_activity_calorie	respiratory_rate	stress
row_0					
dup_0s	1	0	0	13	13
near_dup_0-10s	0	0	0	0	0
high_freq_10s-3min	369699	67936	18861	860006	860681
near5_low_3-<5min	22084	1537	10047	0	0
exact_5min	3115	355	2827	0	0
near5_high_>5-7min	2647	411	3924	0	0
short_gap_7-30min	21424	757	8561	2	2
long_gap_>30min	5	473	3058	4	4

Physical Activity State Intervals

kind	metric	duration	count	pct	value

0	duration	bin	<1	2297.0	1.24	NaN
1	duration	bin	1–5	135749.0	73.41	NaN
2	duration	bin	5–15	30587.0	16.54	NaN
3	duration	bin	15–60	11631.0	6.29	NaN
4	duration	bin	60–240	4636.0	2.51	NaN
5	duration	bin	>240	8.0	0.00	NaN
6	gap	bin	<1	169812.0	100.00	NaN
7	gap	bin	1–5	2.0	0.00	NaN
8	gap	bin	5–15	0.0	0.00	NaN
9	gap	bin	15–60	1.0	0.00	NaN
10	gap	bin	60–240	0.0	0.00	NaN
11	gap	bin	>240	2.0	0.00	NaN
12	overlap	n_intervals				184908
13	overlap	n_with_any_overlap				15055
14	overlap	pct_with_any_overlap				8.141887
15	overlap	n_with_big_overlap				10437
16	overlap	pct_with_big_overlap				5.644429

Sleep Intervals

	kind	metric	bin	count	pct	value
0	duration	bin	<1	0.0	0.00	NaN
1	duration	bin	1–5	999.0	10.12	NaN
2	duration	bin	5–15	3466.0	35.10	NaN
3	duration	bin	15–60	4788.0	48.49	NaN
4	duration	bin	60–240	620.0	6.28	NaN
5	duration	bin	>240	1.0	0.01	NaN
6	gap	bin	<1	6992.0	95.22	NaN
7	gap	bin	1–5	0.0	0.00	NaN
8	gap	bin	5–15	1.0	0.01	NaN
9	gap	bin	15–60	1.0	0.01	NaN
10	gap	bin	60–240	9.0	0.12	NaN
11	gap	bin	>240	340.0	4.63	NaN
12	overlap	n_intervals				9874
13	overlap	n_with_any_overlap				2495
14	overlap	pct_with_any_overlap				25.268382
15	overlap	n_with_big_overlap				2455

16 overlap pct_with_big_overlap 24.863277

	stream	variable	n	n_non_missing	pct_missing	min	p1	median	p99	max
0	glucose	glucose_mg_dl	100444.0	100444.0	0.00	39.00	71.00	123.00	401.00	401.00
1	environment	lch0	6286316.0	6286316.0	0.00	0.00	0.00	0.00	0.04	0.63
2	environment	lch1	6286316.0	6286316.0	0.00	0.00	0.00	0.00	0.10	0.91
3	environment	lch2	6286316.0	6286316.0	0.00	0.00	0.00	0.00	0.15	0.91
4	environment	lch3	6286316.0	6286316.0	0.00	0.00	0.00	0.00	0.20	0.91
5	environment	lch6	6286316.0	6286316.0	0.00	0.00	0.00	0.00	0.36	0.91
6	environment	lch7	6286316.0	6286316.0	0.00	0.00	0.00	0.00	0.58	0.91
7	environment	lch8	6286316.0	6286316.0	0.00	0.00	0.00	0.00	0.49	0.91
8	environment	lch9	6286316.0	6286316.0	0.00	0.00	0.00	0.00	0.33	0.91
9	environment	lch10	6286316.0	6286316.0	0.00	0.00	0.00	0.00	0.91	inf
10	environment	lch11	6286316.0	6286316.0	0.00	0.00	0.00	0.00	0.65	inf
11	environment	pm1	6286316.0	6286316.0	0.00	0.00	0.20	2.50	70.10	3620.50
12	environment	pm2.5	6286316.0	6286316.0	0.00	0.00	0.20	2.70	74.90	3833.50
13	environment	pm4	6286316.0	6286316.0	0.00	0.00	0.20	2.70	76.50	3864.20
14	environment	pm10	6286316.0	6286316.0	0.00	0.00	0.20	2.70	77.50	3879.00
15	environment	hum	6286316.0	6286316.0	0.00	19.37	27.19	43.57	59.78	69.29
16	environment	temp	6286316.0	6286316.0	0.00	12.48	15.93	24.08	31.75	650.37
17	environment	voc	6286316.0	6286316.0	0.00	0.00	16.00	101.00	480.00	500.00
18	environment	nox	6286316.0	6286130.0	0.00	0.00	1.00	1.00	9.00	235.00
19	environment	screen	6286316.0	6286316.0	0.00	0.00	0.00	0.00	0.00	1.00
20	environment	ff	6286316.0	6286316.0	0.00	0.00	0.00	1.00	120.00	120.00
21	environment	inttemp	6286316.0	6286316.0	0.00	13.00	19.75	27.50	35.75	38.00
22	activity_point	heart_rate_bpm	2444032.0	419011.0	82.86	0.00	0.00	76.00	120.00	187.00
23	activity_point	spo2_pct	2444032.0	71448.0	97.08	69.00	81.00	92.00	100.00	101.00
24	activity_point	resp_rate_bpm	2444032.0	860061.0	64.81	-2.00	-2.00	12.51	21.80	31.16
25	activity_point	stress	2444032.0	860736.0	64.78	-2.00	-2.00	-1.00	96.00	100.00
26	activity_point	active_calories_kcal	2444032.0	47314.0	98.06	0.00	1.00	164.00	1113.87	1862.00

	count	mean	std	min	max
lch0	36.0	0.004797	0.008969	0.000009	0.045905
screen	36.0	0.003686	0.012513	0.000038	0.072917
lch1	36.0	0.008218	0.015052	0.000014	0.060867
lch2	36.0	0.011412	0.022147	0.000031	0.095208
lch3	36.0	0.015832	0.029706	0.000042	0.137987

lch6	36.0	0.022746	0.048181	0.000071	0.255418
lch9	36.0	0.026714	0.049262	0.000081	0.215431
lch8	36.0	0.034259	0.070361	0.000083	0.358995
lch7	36.0	0.029801	0.072663	0.000065	0.413892
n_sleep_stages	35.0	3.810907	0.256796	3.000000	4.300000
nox	36.0	1.498051	1.111249	0.989933	5.739332
spo2_pct	35.0	91.481451	1.728232	88.279629	95.556562
Glucose_n_peaks	36.0	9.641288	1.761578	6.000000	14.090909
resp_rate_bpm	35.0	10.879919	2.243524	4.865152	16.129954
temp	36.0	24.241823	2.542338	17.901552	31.481208
inttemp	36.0	27.950160	3.025222	21.575718	35.099967
glucose_cv_pct	36.0	17.820560	4.993786	10.295015	29.770064
glucose_n_readings	36.0	258.807828	5.822817	241.200000	284.900000
hum	36.0	43.429253	6.593935	27.619781	57.196220
pm1	36.0	6.718519	6.690307	0.688385	23.603785
pm2.5	36.0	7.162064	7.139143	0.752655	26.250801
pm4	36.0	7.258623	7.248777	0.778255	27.499433
pm10	36.0	7.305304	7.304595	0.790999	28.112223
n_sleep_intervals	35.0	25.200249	11.280303	12.555556	73.000000
heart_rate_bpm	35.0	76.182539	11.330057	45.075905	96.506783
voc	36.0	122.692078	11.820992	100.322163	141.057973
stress	35.0	32.260463	17.149345	10.214217	77.967214
glucose_time_in_range_70_180_pct	36.0	88.418223	19.806431	0.000000	100.000000
ff	36.0	27.643408	29.275716	0.999855	97.435332
glucose_min	36.0	91.882702	32.588384	57.900000	264.800000
glucose_mg_dl_mean	36.0	135.466660	45.215244	97.720528	371.256413
glucose_max	36.0	203.057639	49.837477	137.818182	401.000000
active_calories_kcal	35.0	209.070083	116.128148	28.377051	500.080981
lch10	36.0	inf	NaN	0.000449	inf
lch11	36.0	inf	NaN	0.000081	inf

Glucose Spike Dataframe:

Columns:

- **person_id:** person id
- **Spike_start_utc:** full timestamp the time spike began
- **Spike_start_date:** calendar day the time spike began

- **Spike_start_time_of_day:** time of day spike began
- **Duration_minutes:** duration of spike (minutes)
- **Mean_blood_sugar_mg_dl:** mean blood sugar during spike (mg/dl)
- **Max_blood_sugar_mg_dl:** max blood sugar during spike (mg/dl)

93,316 rows across 2245 patients

Spikes with mean <= 140: 0

Spikes with max <= 140: 0

Negative durations

USED LINEAR INTERPOLATION FOR ADJACENT POINTS TO ADDRESS SINGLE POINT SPIKES

Overlapping Physical Activity State Intervals

Pair (prev | curr)	Count	% of overlaps
sedentary | NA	6,134	40.74%
NA | sedentary	5,725	38.03%
NA | NA	2,430	16.14%
NA | walking	317	2.11%
walking | NA	312	2.07%
NA | generic	77	0.51%
generic | NA	60	0.40%

```
pair_signature,n_rows,pair_bucket,pct_of_overlaps
light|light,1164,both_ordered_same_stage,46.6533
awake|awake,475,both_ordered_same_stage,19.0381
deep|deep,430,both_ordered_same_stage,17.2345
rem|rem,362,both_ordered_same_stage,14.509
awake|light,15,both_ordered_adjacent_stages,0.6012
deep|light,12,both_ordered_adjacent_stages,0.481
light|deep,12,both_ordered_adjacent_stages,0.481
light|rem,6,both_ordered_two_steps_apart,0.2405
rem|light,5,both_ordered_two_steps_apart,0.2004
light|awake,5,both_ordered_adjacent_stages,0.2004
deep|awake,4,both_ordered_two_steps_apart,0.1603
rem|deep,2,both_ordered_adjacent_stages,0.0802
awake|deep,1,both_ordered_two_steps_apart,0.0401
awake|rem,1,both_ordered_three_plus_steps_apart,0.0401
deep|rem,1,both_ordered_adjacent_stages,0.0401
```

Daily Patient Spike Aggregates Dataframe (spike_daily_by_cgm_day):

- Associated the spike to a day only if startdate occurs in it instead of associating spikes to days if the spike occurs in it at all and splitting if occur in 2 different days
- One row per person-day with at least 1 CGM reading
- Columns: person_id, date, mean_duration_minutes, n_spikes
- Coverage: 24,012 person-days total; 21,023 days with >=1 spike and 2,989 days with 0 spikes
- Zero-spike handling: days with n_spikes = 0 are retained, and mean_duration_minutes is NaN on those days

4/29/2026:

Patient Spike Aggregates Dataframe (spike_aggregates_by_person):

- Columns: person_id, avg_cgm_per_spike_mg_dl, avg_spike_duration_minutes, avg_spikes_per_day, total_spikes, days_tracked
- Rows: 2,245
- {Days tracked: frequency} 2: 8, 3: 5, 4: 6, 5: 9, 6: 11, 7: 20, 8: 25, 9: 47, 10: 190, 11: 1913, 12: 6, 13: 3, 14: 1, 15: 1
    - 85% of participants have 11 tracked days

Added linear interpolation to tracking spikes:

- For each detected spike:
    - Start boundary: between last <=140 point and first >140 point
    - End boundary: between last >140 point and next <=140 point
- Estimate the exact timestamp that crosses 140 mg with linear interpolation between those two points as the actual start and end of the spike
- Spike duration = interpolated_end - interpolated_start
- Eliminates 1 timestamp spikes that would be have to imputed with 0 or 5 minutes

Spearman Correlation Test for features with MOCA score:

predictor	n	spearman_rho	p_value	p_value_fdr_bh	significant_fdr_0_05
meta_age	2254	-0.187904	2.334391e-19	1.634074e-18	True
mean_glucose	2233	-0.173571	1.454253e-16	5.089886e-16	True
total_movement	2107	0.075280	5.434982e-04	1.268163e-03	True
day_hr	1988	-0.072836	1.154896e-03	1.765488e-03	True
night_mean_hr	1988	-0.072275	1.261063e-03	1.765488e-03	True
night_movement	2107	0.064048	3.269435e-03	3.814341e-03	True
min_spo2	1620	0.026033	2.950131e-01	2.950131e-01	False